

Reproduction of linkedin/Liger-Kernel: fused linear cross entropy, RMSNorm and SwiGLU benchmarks

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Independent reproduction · Artifact evaluation · Systems and performance

Abstract

Ran three of the repository's kernel benchmarks at the pinned commit (fused linear cross entropy, RMSNorm, SwiGLU) on Triton 3.6 with PyTorch 2.11, comparing Liger's kernels with the baseline each script uses (Hugging Face modules, or plain PyTorch for fused linear cross entropy) in speed and peak memory, forward, backward and full passes. The report holds the geometric-mean ratio of baseline to Liger over each script's sweep; values above 1 favour Liger. On this GPU the fused linear cross entropy kernel uses less memory but is slower than the PyTorch baseline. One NVIDIA RTX 5060 Ti (Blackwell, sm_120).

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	FLCE, RMSNorm, SwiGLU v...	2026-10-10	1	1	Signed by the K-Veritas server

2 Results

r1: FLCE, RMSNorm, SwiGLU vs baseline

Run	Metric	Final value	Step
1	fused_linear_cross_entropy_backward_speed_ba...	5.34787	
1	fused_linear_cross_entropy_forward_speed_base...	0.0909073	
1	fused_linear_cross_entropy_full_memory_baseli...	1.23818	
1	fused_linear_cross_entropy_full_speed_baseline_...	0.275131	
1	rms_norm_backward_speed_baseline_over_liger	15.0434	
1	rms_norm_forward_speed_baseline_over_liger	7.91171	
1	rms_norm_full_memory_baseline_over_liger	4.70807	
1	rms_norm_full_speed_baseline_over_liger	8.96247	
1	swiglu_backward_memory_baseline_over_liger	1.12276	
1	swiglu_backward_speed_baseline_over_liger	1.01894	
1	swiglu_forward_memory_baseline_over_liger	1.1035	

Run	Metric	Final value	Step
1	swiglu_forward_speed_baseline_over_liger	1.01625	
1	swiglu_full_memory_baseline_over_liger	1.12276	
1	swiglu_full_speed_baseline_over_liger	1.01778	
1	rows_measured	92	

Run	Inline claim	Value	Phase
1	fused_linear_cross_entropy_full_memory_baseli...	1.23818	
1	fused_linear_cross_entropy_full_speed_baseline_...	0.275131	
1	rms_norm_full_memory_baseline_over_liger	4.70807	
1	rms_norm_full_speed_baseline_over_liger	8.96247	
1	swiglu_full_memory_baseline_over_liger	1.12276	
1	swiglu_full_speed_baseline_over_liger	1.01778	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	202.0 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	0257a92ed6b937f09c916367069b7740742b1c07065101486610a24a1e24a235	2.2 MB
r1.kvbundle.zip	6ab2beb6c789f698104452416e4907dda8c32e277b63a7fea484b3285a475c33	1.6 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2410.10989>

Reproduction of: <https://github.com/linkedin/Liger-Kernel/tree/d5f281789aeb5d8d13ffb6825d8e74486af7442c>

Record: CC-BY 4.0. Code: BSD-2-Clause, as declared by the authors. Tags: triton, gpu-kernels, llm-training, benchmark.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (FLCE, RMSNorm, SwiGLU vs baseline), kv:2610.00035v1/r1. Data hash 8e2bf2961fc0da8e721b6f9c0eec4508e92617114d33f4893e7cd6d64efeeadc.
<https://kveritas.org/records/2610.00035v1/r1.pdf>